Beyond Pixel Overlap: A Structural and Topological Audit of Dice Limitations in Retinal Vessel Segmentation

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IEEE Latin American Conference on Computational Intelligence – LA-CCI 2026
Lima, Peru · November 2026

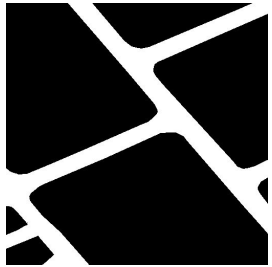
Thin Network-Like Structures

Tubular, network-like structures are relevant to many fields (Shit et al., CVPR 2021)

Roads · remote sensing



image



ground truth

A gap breaks the route network.

RoadNet dataset (Ottawa) – Liu et al., IEEE TGRS, 2019

Cracks · structural inspection



image

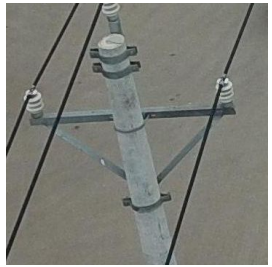


ground truth

A gap splits one crack into several.

DeepCrack dataset – Liu et al., Neurocomputing, 2019

Power lines · grid inspection



image



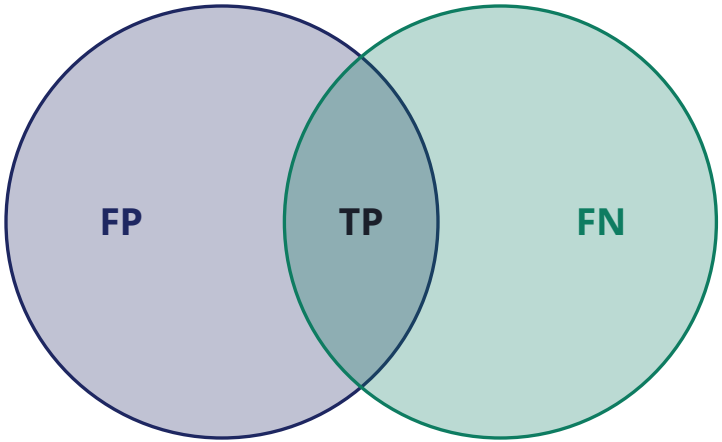
ground truth

A gap breaks the power line into pieces.

PLD-UAV dataset (PLDU) – Zhang et al., Remote Sensing, 2019

Thin structures must stay connected and missing pixels can break the network that the application depends on.

How Segmentation Is Usually Evaluated: Dice and IoU



TP = Ŷ ∩ Y · FP = Ŷ only · FN = Y only

Dice coefficient

$$\mathrm{Dice}(\hat{Y}, Y) = \frac{2\,|\hat{Y} \cap Y|}{|\hat{Y}| + |Y|} = \frac{2\,TP}{2\,TP + FP + FN}$$

Intersection over Union (IoU) · Dice = 2·IoU / (1 + IoU)

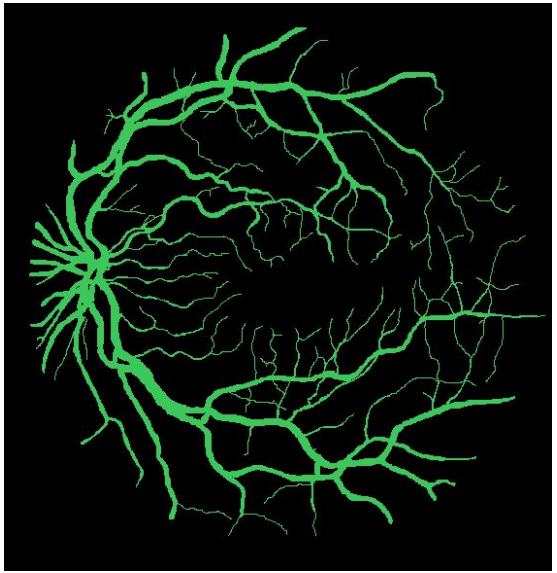
$$\mathrm{IoU}(\hat{Y}, Y) = \frac{|\hat{Y} \cap Y|}{|\hat{Y} \cup Y|} = \frac{TP}{TP + FP + FN}$$

Dice loss (training)

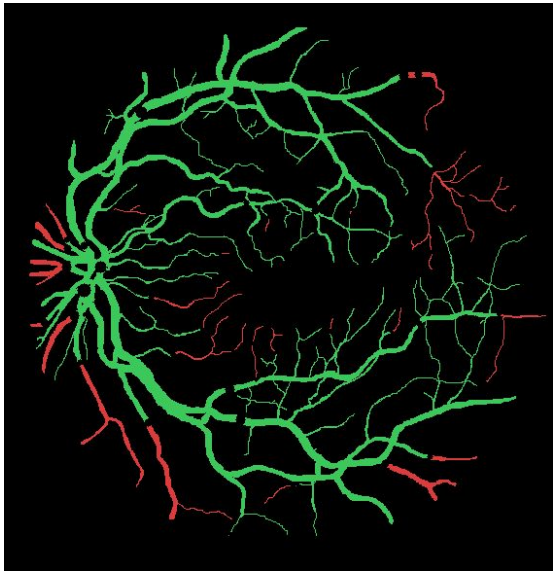
$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_i \hat{y}_i\, y_i + \varepsilon}{\sum_i \hat{y}_i^2 + \sum_i y_i^2 + \varepsilon}$$

Both metrics and the loss function only count overlapping pixels. None of them checks whether the structure stays connected.

The Topological Blindness of Dice



(a) Ground-truth mask



(b) Fragmented version of (a)

0.9814

Dice of (b) against (a)

33

connected components in (b) – red: cut off from the main tree

Near-perfect overlap (high Dice), yet the vessel tree is split into 33 pieces. An example of a topological error that Dice cannot see.

Motivation: Black-Box Disease Classification

The retina offers a direct view of the microcirculation: vascular alterations are early indicators of diabetic retinopathy, hypertension and cardiovascular disease.



Retinography

End-to-end classifier

Output: a single label

- Disease
- No disease

? Which vessels or structures drove the decision?

? How can relevant features be identified from the model alone?

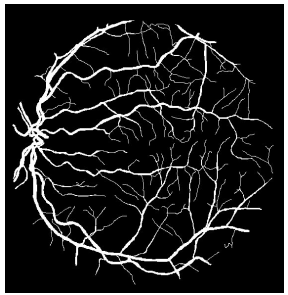
? How can a clinician verify or monitor the result over time?

End-to-end classifiers return a label, not an explanation. Working as a black-box poses challenges for clinical adoption.

Motivation: Segmentation Enables Structural Analysis



Retinography



Vessel segmentation

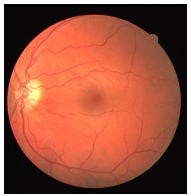
Measurable structure example

- Vessel caliber
- Branching points and junctions
- Tortuosity and length
- Network connectivity

- **From labels to biomarkers:** interpretable (XAI) pipelines extract structural biomarkers from the vessel tree instead of predicting a single label.
- **Segmentation is the prerequisite:** every structural biomarker is computed on the segmented vessel tree.

From Fundus Images to Structural Biomarkers

Hybrid framework for quantifying retinal vessel networks proposed by our group (Silva et al., 2025)



Retinography

1

Segmentation

Processes the retinography and extracts the vessels.

▲ audited in this work

2

Rendering

Converts the segmented image into surface meshes.

3

Graph Generation

Centerlines → topological and semantic graphs.

4

Network Analysis

Network metrics (degree, clustering, centralities) and GNN encoding.

5

Qualitative Inference

Patterns of disease, anomalies, aging or neovascularization.

Every downstream biomarker inherits the segmentation: a single missing pixel at a vessel connection splits the tree, adds a branch and changes the network metrics.
Goal: early detection of systemic and ocular diseases from the retinal vascular structure.

H. Silva et al., “A Hybrid Framework for Quantifying and Analyzing the Structural Properties of Human Retinal Vessel Networks,” Complex Networks XVI, 2025.

The Evaluation Gap

Pixel-wise metrics still dominate

- Recent architectures (DR-VNet, SDDC-Net, RetinaLiteNet, LVS-Net, LFRA-Net, LFA-Net) are reported with Sensitivity, Specificity, Accuracy and Dice.
- These measure spatial overlap – not the topology of the vascular tree.
- A model can reach a **high Dice** while producing **disconnected vessels** or wrong junctions.

Topology-aware losses

- **clDice** (Shit et al., CVPR 2021) penalizes topological breaks by comparing skeletons of prediction and reference.
- **cbDice** (Shi et al., MICCAI 2024) adds the Euclidean Distance Transform to handle vessel-radius variation.
- Used as a **metric**, clDice is still a skeletal overlap.

We propose a multidimensional audit that quantifies the structural errors pixel-wise metrics miss.

Experimental Design

1 Backbone: LFA-Net at native resolution

- Lightweight, efficient vessel-segmentation network.
- Original 512×512 resize removed – its interpolation breaks vessels.
- Padding 565×584 → 568×584; output cropped back for pixel-perfect alignment.

2 Data: DRIVE

- Train split: 16 training + 4 validation images; 1st annotator as ground truth.
- CLAHE + augmentation → 320 samples (geometric, photometric, noise, elastic/optical distortions, coarse dropout).
- 20 test images for every evaluation.

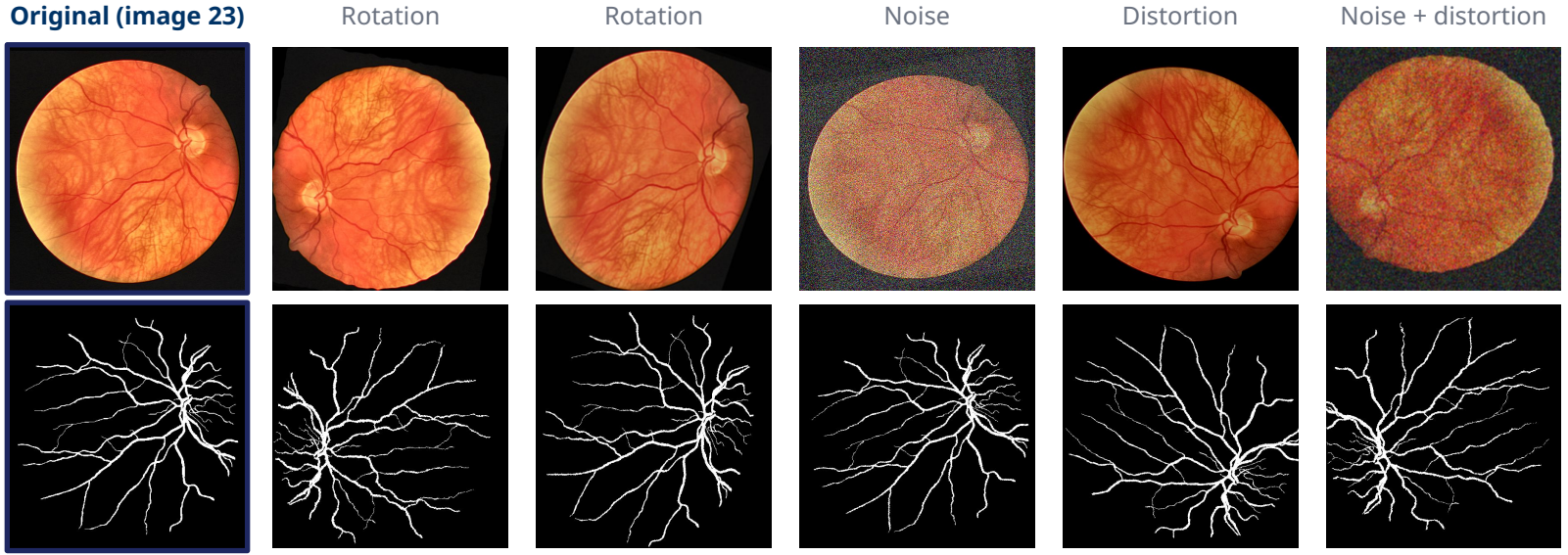
3 Training

- Adam, initial learning rate 0.001, batch size 16.
- 300 fixed epochs + 300 epochs with LR × 0.5 after 50 epochs of loss stagnation.
- Early-stopping patience: 200 epochs.

4 Loss functions

- **Dice Loss (DL)** · **Weighted Dice Loss (WDL)** · **clDice** · **cbDice**
- Every prediction audited **raw** and **filtered** (small disconnected components removed, no reconnection).

Training Data: DRIVE Augmentation



Top: retinography · Bottom: ground-truth mask · Geometric transforms are applied to both

16 × 20 = 320

training images × augmentations = samples

Geometric: flips, shifts, scales, rotations **Photometric:** brightness, contrast, gamma
Noise: Gaussian **Distortions:** elastic, optical **Coarse dropout**

A Multidimensional Evaluation Framework

1

Structural complexity

Inherent topological difficulty of each image

- **TTVR** (proposed): Thin-to-Thick Vessel Ratio.
- Thin vessels: 1 pixel wide.
- Higher TTVR → more microvasculature → harder to preserve continuity.

2

Pixel-wise overlap

Global and microvascular overlap

- Sensitivity (recall), Specificity, Dice.
- **ThinVesselHit**: recall only on vessels ≤ 2 px wide.
- **FP_Rate** = FP / (TP + FP), raw vs. filtered: background noise vs. thickening attached to the tree.

3

Topological audit

Structural connectivity

- **clDice** (as a metric): centerline consistency.
- **Betti errors**: β0 = structural breaks; β1 = false loops / incorrect bridges.

4

Network science

Functional integrity of the vessel graph

- Skeleton → graph G = (V, E).
- **Generated components**: global fragmentation.
- On the **Largest Connected Component (LCC)**: nodes, edges, endpoints, junctions, average degree.

Topological Complexity Profiling (TTVR)

Img.	TTVR	Img.	TTVR
1	0.781	11	1.273
2	0.716	12	0.985
3	0.968	13	1.170
4	1.047	14	1.031
5	0.878	15	0.950
6	1.020	16	0.592
7	0.807	17	0.586
8	0.873	18	0.542
9	1.057	19	0.759
10	1.109	20	0.607

Table I (TTVR) · red: dense capillaries · green: lowest TTVR

Higher TTVR — More 1-pixel-wide vessels — Harder to preserve connectivity

Pearson r · 20 test images · averaged over the four losses

−0.70

clDice ↓

centerline overlap degrades

+0.73

β1 error ↑

more false loops / incorrect bridges

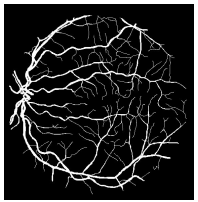


Image 11
TTVR 1.273
harder

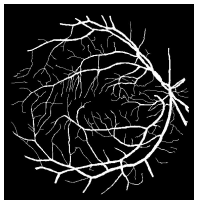


Image 13
TTVR 1.170
harder

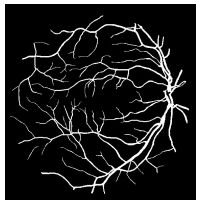
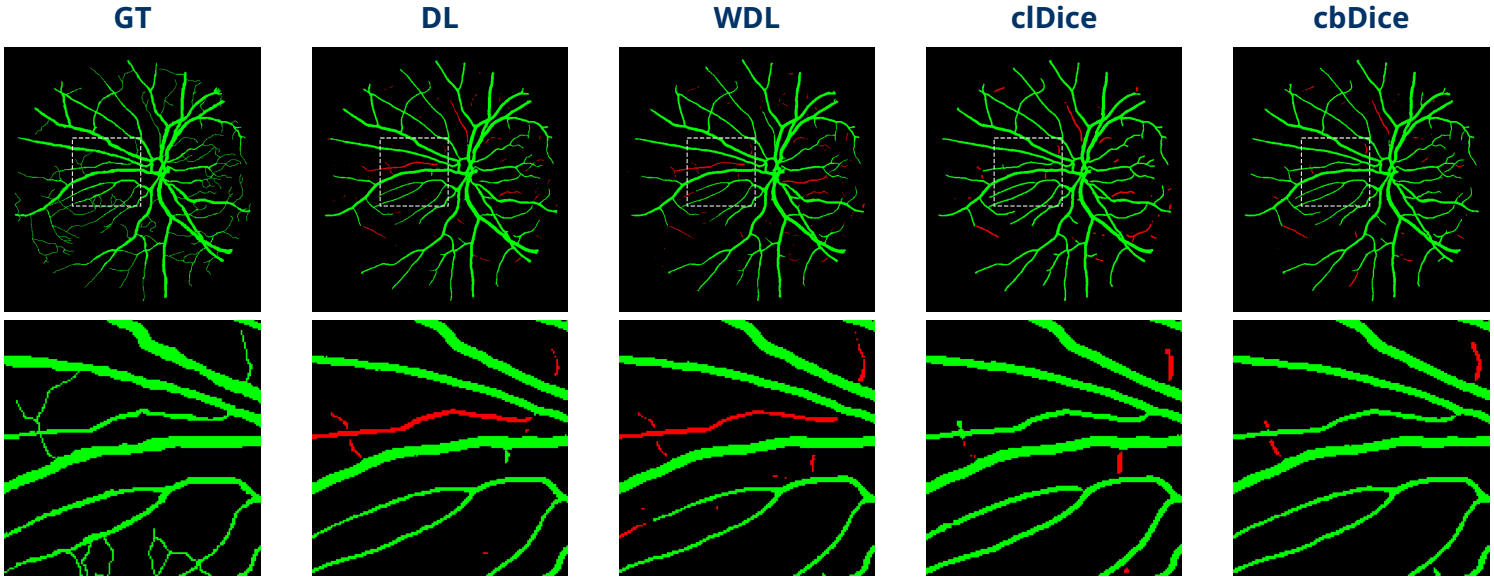


Image 18
TTVR 0.542
easier

Qualitative Audit of Connectivity (Image 4)



Largest connected component Disconnected segments Bottom row: zoom on the dashed region

Topology-aware losses (**clDice**, **cbDice**) maintain continuity in regions where the standard models (DL, WDL) break the vessel.

Raw Predictions: Dice Is Nearly Invariant

Metric	Dice Loss	Weighted DL	clDice Loss	cbDice Loss
Dice ↑	**0.8085 ± 0.0154**	0.8084 ± 0.0170	0.7875 ± 0.0161	0.8068 ± 0.0147
clDice ↑	0.8061 ± 0.0396	0.8000 ± 0.0408	**0.8347 ± 0.0379**	0.8266 ± 0.0399
Sensitivity ↑	0.8040 ± 0.0538	0.7936 ± 0.0547	**0.8502 ± 0.0452**	0.8238 ± 0.0512
Specificity ↑	0.9828 ± 0.0051	**0.9842 ± 0.0046**	0.9708 ± 0.0059	0.9795 ± 0.0048
ThinVesselHit % ↑	49.72 ± 12.16	48.42 ± 11.69	**60.36 ± 11.86**	58.48 ± 12.12
FP_Rate % ↓	18.15 ± 4.51	**17.11 ± 4.09**	26.30 ± 4.30	20.53 ± 4.02
Betti-0 error ↓	115.3 ± 19.72	132.0 ± 21.05	**46.9 ± 10.91**	72.65 ± 14.90
Betti-1 error ↓	36.70 ± 16.31	36.95 ± 17.06	**33.55 ± 17.19**	35.55 ± 17.38
Betti total ↓	152.0 ± 28.25	168.95 ± 32.31	**80.45 ± 21.47**	108.2 ± 27.10
Gen. components ↓	89.2 ± 14.32	97.4 ± 13.12	**43.7 ± 8.83**	63.2 ± 12.86

best worst Table II · mean ± SD, 20 test images · ↑ higher / ↓ lower is better

Reading the table

- **Dice:** 0.787–0.808 – by Dice alone, the four models look nearly identical.
- **clDice metric:** best for clDice loss (0.834), a marginal gain – still a skeletal overlap.
- **Components:** 89.2 (DL) and 97.4 (WDL) vs. 43.7 (clDice) and 63.2 (cbDice).
- **β0 error ≠ components:** computed differently – Betti Error metric vs. count on the skeleton graph.

Dice barely changes across losses. Metrics such as **ThinVesselHit**, **Betti errors** and **generated components** highlight differences between the models.

Network-Science Audit - Largest Connected Component

Network metric	Ground Truth	DL	WDL	clDice	cbDice
Nodes	**344.90 ± 75.07**	179.70 ± 31.98	174.50 ± 30.63	**199.30 ± 32.18**	197.00 ± 36.82
Edges	**389.65 ± 85.53**	198.25 ± 37.00	192.95 ± 35.67	**220.90 ± 36.91**	217.20 ± 41.91
Endpoints (degree 1)	**125.30 ± 26.69**	70.60 ± 10.95	68.20 ± 11.08	77.25 ± 12.18	**77.60 ± 13.54**
Junctions (degree ≥ 3)	**213.00 ± 46.56**	107.30 ± 20.58	104.75 ± 20.58	**120.05 ± 20.84**	117.70 ± 23.29
Average degree	**2.26 ± 0.04**	2.20 ± 0.03	2.21 ± 0.04	**2.22 ± 0.04**	2.20 ± 0.03

Table IV · vessels skeletonized into a graph; metrics on the LCC only (disconnected fragments ignored) · mean ± SD

LCC metrics were identical for raw and filtered predictions. The filter removes only isolated noise and not the main vascular backbone (largest connected component).

Conclusions

1

Dice is structurally blind

Near-identical overlap (**0.787–0.808**) hides vascular trees fragmented into **43.7 to 97.4** generated components (raw predictions).

2

Topology-aware losses carry a trade-off

clDice improves continuity and thin-vessel recall, but **dilates vessels** – distorting caliber, a primary biomarker for cardiovascular and hypertensive risk.

3

Evaluation must go beyond overlap

Clinical reliability requires structural auditing that demands both **topological continuity** and **morphological fidelity**.

Limitations and Future Work

Scope and limitations

- Limited to **one architecture and one dataset** (LFA-Net, DRIVE): the contribution is the evaluation framework, so the effect of the loss function is isolated.
- **Preliminary check:** predictions from four additional architectures, examined with the same network-science metrics, showed consistent topological deficiencies.

Future work

Since this paper: the master's dissertation extended the audit to LVS-Net and DR-VNet trained with the four losses. The behavior was consistent: topology-aware losses reduced fragmentation, while Dice remained insensitive to topological integrity.

R. Albuquerque, M.Sc. dissertation, PPGEC/UPE, 2026.

- Apply the framework to **other retinal datasets** and **larger models**, such as Transformers.
- Evaluate more recent loss functions and propose new network-science descriptors.

Segmentation cannot be judged by spatial overlap alone. Structural audits should guide new loss functions and metrics that preserve both topological continuity and morphological fidelity.

Thank you!

Questions?

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Schedule a meet:
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How TTVR and the Vessel Graph Are Computed

Thin-to-Thick Vessel Ratio (TTVR)

$$TTVR = N_{thin} / N_{thick}$$

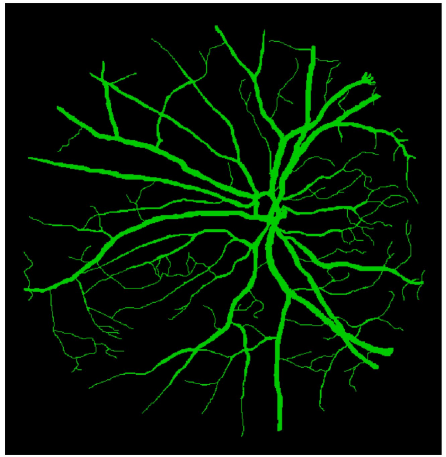
- Vessel width from the **morphological skeleton** + **Euclidean Distance Transform (EDT)**.
- **Thin:** width = 1 px, the scale most susceptible to connectivity breaks.
- **Thick:** width > 1 px.
- Higher TTVR → more microvasculature → harder to preserve continuity.
- DRIVE test set: **0.542** (image 18) to **1.273** (image 11).

Vessel graph (network science)

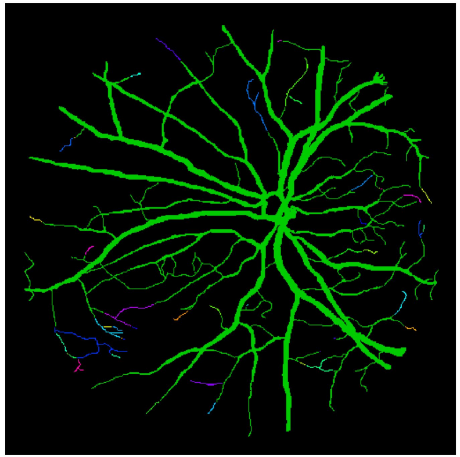
Binary mask	prediction or ground truth
↓	
Skeleton	thinning algorithm of **Lee et al. (1994)** – 1-px centerlines
↓	
Graph G = (V, E)	skeleton converted into a graph; descriptors computed with **NetworkX**
↓	
Metrics	**Components:** whole graph. **LCC only:** nodes, edges, endpoints (degree 1), junctions (degree ≥ 3), average degree

Lee, T. C.; Kashyap, R. L.; Chu, C. N. Building skeleton models via 3-D medial surface axis thinning algorithms. CVGIP: Graphical Models and Image Processing, 56(6), 1994.

Resizing Fragments the Ground Truth



(a) Image 4, original GT – 1 component



(b) 512×512 resized GT – 47 components

Resizing is a common pre-processing step to fit model architectures or to combine datasets with different native resolutions.

Img.	GT	Resize	Img.	GT	Resize
1	7	37	**11**	3	59
2	3	33	**12**	1	42
3	1	38	**13**	5	36
4	**1**	**47**	**14**	1	37
5	1	46	**15**	2	28
6	4	41	**16**	2	7
7	2	34	**17**	1	12
8	3	32	**18**	3	13
9	3	28	**19**	5	24
10	3	46	**20**	3	10

Table I · connected components before / after resize · red: image 4

Resizing increased the number of components in all 20 test images – interpolation breaks the native connectivity, so this study uses padding and cropping at native resolution.

Post-Processed Predictions

Metric	Dice Loss	Weighted DL	clDice Loss	cbDice Loss
Dice ↑	0.8058 ± 0.0162	0.8052 ± 0.0187	0.7903 ± 0.0155	**0.8083 ± 0.0147**
clDice ↑	0.7960 ± 0.0451	0.7898 ± 0.0458	**0.8339 ± 0.0403**	0.8241 ± 0.0438
Sensitivity ↑	0.7887 ± 0.0559	0.7776 ± 0.0575	**0.8448 ± 0.0457**	0.8149 ± 0.0522
Specificity ↑	0.9843 ± 0.0049	**0.9857 ± 0.0045**	0.9723 ± 0.0059	0.9811 ± 0.0047
ThinVesselHit % ↑	42.89 ± 12.70	41.13 ± 12.02	**57.91 ± 12.28**	54.08 ± 12.69
FP_Rate % ↓	17.07 ± 4.47	**15.94 ± 4.13**	25.39 ± 4.29	19.38 ± 3.97
Betti-0 error ↓	**5.1 ± 2.69**	5.15 ± 2.98	9.70 ± 3.96	7.60 ± 3.80
Betti-1 error ↓	36.75 ± 16.35	36.95 ± 17.06	**33.90 ± 17.08**	35.55 ± 17.38
Betti total ↓	**41.85 ± 16.64**	42.10 ± 17.08	43.60 ± 19.12	43.15 ± 18.52
Gen. components ↓	**8.0 ± 2.43**	8.05 ± 2.89	12.60 ± 3.39	10.50 ± 3.35

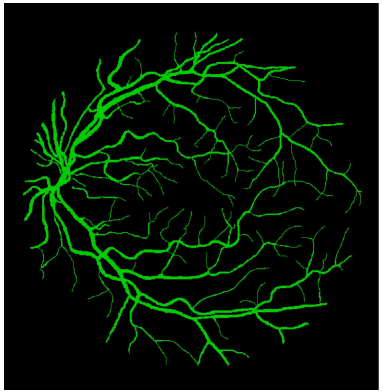
best worst Table III · mean ± SD, 20 test images · ↑ higher / ↓ lower is better

Reading the table

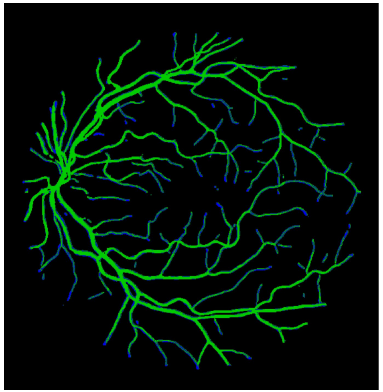
- **Filter:** removes small disconnected components without reconnecting vascular branches.
- **Fewer components ≠ better connectivity:** DL now has the fewest (8.0 vs. 12.6), yet the main tree (LCC) is still larger with clDice (Table IV).
- **clDice loss** keeps the best clDice, Sensitivity and ThinVesselHit.

FP_Rate barely changes after filtering (DL 18.15% → 17.07%; clDice 26.30% → 25.39%) – false positives are not isolated noise.

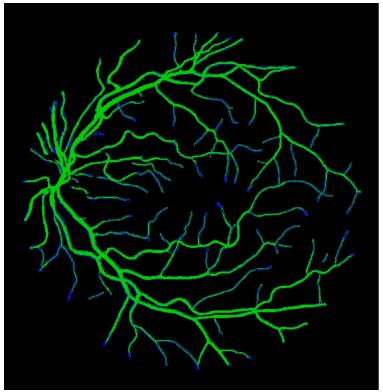
Vessel Thickening



(a) GT



(b) Raw prediction



(c) Filtered prediction

FP_Rate: raw → filtered

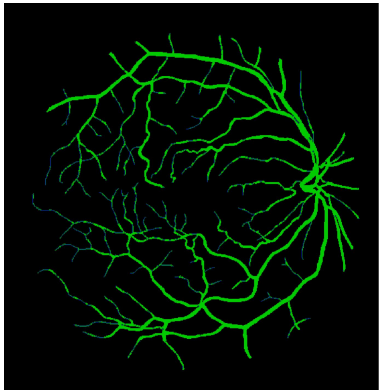
Image 1, clDice Loss: 31.30% → 30.80%

If false positives were isolated noise, the filter would cut FP_Rate sharply.

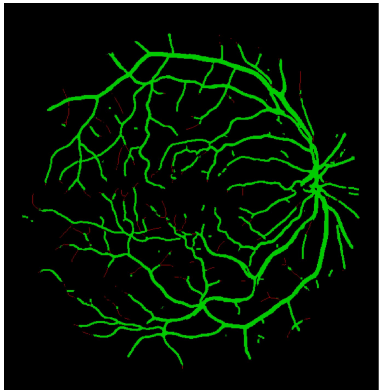
- **False positives (blue) are structural extensions** of the vessel tree – thickened borders, not background artifacts. Part of them comes from recovering more thin vessels.
- **Topology-aware losses dilate vessels** to create a structural “safety margin” that minimizes disconnection penalties.

Dilation may distort vessel caliber, which is a biomarker for cardiovascular risk and hypertension, and classic metrics do not highlight this.

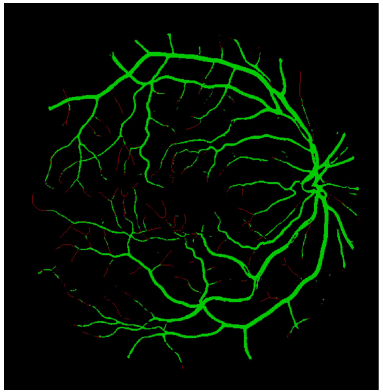
ThinVesselHit



(a) GT – vessels ≤ 2 px in blue



(b) clDice loss – 61.19%



(c) WDL loss – 40.71%

ThinVesselHit

Recall on vessels ≤ 2 px wide

Mean, 20 test images
DL: 49.72%
WDL: 48.42%
clDice: **60.36%**
cbDice: 58.48%

Image 17 (shown left)

- **DL and WDL** prioritize the thick, prominent vessels, which dominate the error gradient, and systematically miss 1-to-2-pixel vessels. Missed vessels in red (image 17).
- **Topology-aware losses force the network to segment microstructures,** which are important to maintain the global skeleton.